

Eradication of Norway Rats for Recovery of Seabird Habitat on Langara Island, British Columbia

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Abstract

Introduced *Rattus norvegicus* (Norway rats) caused the decline of *Synthliboramphus antiquus* (ancient murrelets) and other seabirds breeding on Langara Island (approximately 3,100 ha), British Columbia. Using funds from the litigation settlement following the *Nestucca* oil spill, Environment Canada eradicated Norway rats using a technique developed in New Zealand which involved dispensing wax baits containing the anticoagulant brodifacoum at 50 ppm from fixed bait stations. Bait stations were placed every 75 to 100 m on a grid over the entire island (1 station/ha). Rats removed bait for 26 days, after which crews placed baits in protective plastic bags in each bait station. Stations loaded with baits were left on the island and rechecked four times over 2 years, after which bait stations and remaining bait were removed. The eradication succeeded. No signs of rats have been detected on Langara Island and its associated islands since January 1996. No rats were trapped during 1,700 trap-nights following the poison campaign. Incisor marks of rats were not found on apples or oil-dipped chew-sticks. *Corvus corax* (common ravens) likely suffered greater than 50% mortality from the eradication after apparently gaining access to the poison directly from bait stations and from scavenging rat carcasses. A monitoring and response

system is being developed in conjunction with current users of the islands. The success on Langara Island demonstrates how the technique proven on small New Zealand islands of less than 300 ha can be effectively extrapolated to much larger islands.

Key words: seabirds, habitat recovery, Langara Island, rat eradication, brodifacoum, *Rattus norvegicus*, Norway rat, British Columbia, non-target poisoning.

Introduction

Langara Island (54°15'N, 133°01'W), at the northwest tip of the Queen Charlotte Islands (Haida Gwaii), British Columbia, Canada, offers a classic example of seabird decline for the temperate region of the world (Bailey & Kaiser 1993; Bertram 1995). The island was once one of British Columbia's largest seabird colonies, with six species of burrow-nesting seabirds (Drent & Guiguet 1961). Between 1971 and 1980, the island lost its breeding populations of *Ptychoramphus aleutica* (Cassin's auklets), *Cerorhinca monocerata* (rhinoceros auklets), *Oceanodroma furcata* (fork-tailed storm petrels), and *O. leucorhoa* (Leach's storm petrels), and scattered colonies of *Fratercula cirrhata* (tufted puffin) were reduced to a single site atop Cox Island (Vermeer & Sealy 1984).

The island's ancient murrelet (*Synthliboramphus antiquus*) population once described as "immense" and "astronomical" (Beebe 1960), and estimated at 200,000 pairs (Gaston 1992), had declined both in numbers and extent of nesting area since the 1950s. In 1971 the population had shrunk to an estimated 80,000 to 90,000 breeding pairs (Vermeer & Sealy 1984), and by 1988 to about 24,000 pairs (Bertram 1989, 1995). During the next 5 years the murrelet colony decreased further to about 14,600 pairs (Harfenist 1994).

The Langara Island Seabird Habitat Recovery Project (LISHRP) began in 1988 with the realization that introduced *Rattus* spp. (rats) had caused this drastic decline and local extinction of breeding seabirds on Langara Island (Bertram 1989, 1995; Kaiser et al. 1997). Although introduced rats were known predators of ancient murrelets, they were long overlooked as a cause of the bird's declining numbers on Langara Island. Avian predators, competition from salmon fisheries, or oceanographic changes were thought to be mainly responsible (Vermeer et al. 1984). Bertram (1989, 1995) finally identified rats as a major factor in the decline when he found dead adult ancient murrelets with wounds at their napes, typical of rat predation on seabirds (Moller 1983). In addition, he found murrelet bones in almost 30% of breeding murrelet burrows. Skeletal remains were most common in abandoned areas of the seabird colony and were least common in areas where burrows remained occupied (Bertram 1995).

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Rats have devastated seabird populations throughout the world (Moors & Atkinson 1984; Atkinson 1985), although this was the first time they were recognized as a significant problem in Canada (Bailey & Kaiser 1993; Harfenist 1994). Of the three widespread commensal rat species, the large ground-dwelling *Rattus norvegicus* (Norway rat) has by far the greatest impact on colonial nesting seabirds (Imber 1975; Norman 1975). The earliest record of rats on Langara Island was of *Rattus rattus* (black rats) in 1946. Norway rats displaced black rats from Langara Island during the period of the seabird decline (Bertram & Nagorsen 1995), and dispersed to nearby Cox and Lucy islands. The exact mechanism of introduction for rats is unknown, although light station construction and maintenance, the military presence during World War II, barge traffic for forestry and mining operations, and intense salmon fishing in the 1950s and 1960s all included activities that could have introduced rats (Bertram & Nagorsen 1995).

Planning for conservation management did not begin until 1992 (Kaiser et al. 1997) because none of the previously proposed solutions was permanent and required a commitment to repeated expenditures. However, a successful technique for clearing rats from islands had been developed and tested in New Zealand in 1988 (Taylor & Thomas 1989, 1993). This technique involved dispensing poison bait from fixed stations spaced such that at least one station lay within the home range of each rat.

The first reports of this success reached Environment Canada during assessment of damages from the *Nestucca* oil spill, which had killed some 50,000 wintering seabirds along the southwest coast of Vancouver Island (Burger 1993a, 1993b). Rat eradication was proposed as a way of mitigating that loss, through restoration of island habitat for a similar number of birds. It was believed that Langara Island would again become a major seabird-breeding site if rats were removed (Kaiser et al. 1997). Seabird species now extinct on Langara Island bred on other nearby islands without rats. Furthermore, Langara Island still retained relict colonies of puffins and ancient murrelets and provided ideal terrestrial and marine habitat (Kaiser & Forbes 1992).

This paper outlines the rat eradication campaign on Langara Island, which involved four basic phases: (1) planning and environmental review, (2) Lucy Island pilot project, (3) the main eradication campaign in 1995, and (4) follow-up and monitoring. The success of this eradication demonstrates how the technique proven on small New Zealand islands of less than 300 ha was effectively extrapolated to a much larger island.

Study Area

Langara Island (3,105 ha) is the largest of the Kiisgwaii Archipelago near the northwest tip of the Queen Char-

lotte Islands (Haida Gwaii) (Fig. 1). The other islands in the group are Lucy (37 ha), Cox (11 ha), several vegetated islets and stacks, and a few off-shore rocks. All are less than 300 m from Langara Island and within the swimming range for Norway rats. The Archipelago is separated from Graham Island by Parry Passage, a 1 km stretch of strong tidal currents that rats are unlikely to cross without human assistance (Taylor 1984).

Langara Island has a low-lying, undulating topography (160 m elevation). Its west and north coasts are rocky, with steep cliffs, and deeply dissected by steep ravines, whereas the east and southwest coasts are gently sloping and include sand and cobble beaches. Lucy Island is low-lying (69 m elevation) and completely forest covered. Cox Island is a towering rock with coastal forest covering its summit (114 m elevation) and eastern slopes. All of the other islets and stacks with permanent vegetation were included in the project area.

The three main islands have coastal forest of *Picea sitchensis* (Sitka spruce) with an understory of *Calamagrostis nutkaensis* (Nootka reed grass). In addition, Langara and Lucy islands have inner stands of *Thuja plicata* (western redcedar) and *Tsuga heterophylla* (western hemlock), with understories of moss or *Gaultheria shallon* (salal). Langara Island has several lakes and permanent streams, extensive sphagnum bogs, dense thickets of tall salal, and patches of *Calamagrostis* and *Elymus* grasses, particularly on headlands and old native village sites (Vermeer et al. 1984; Rodway et al. 1994).

Native mammals of Kiisgwaii include *Sorex monticolus* (dusky shrew), *Myotis lucifugus* (little brown bat), *Lutra canadensis* (river otter), and at least historically, *Peromyscus maniculatus* (deer mouse). Except for a solitary deer mouse found in August 1994, none has been seen on Kiisgwaii since 1946.

The islands are remote and receive little human traffic for much of the year. The Canadian Coast Guard maintains the Langara Point lightstation, and fishing lodges operate recreational fisheries out of Henslung Cove between May and October each year. Henslung Cove also provides safe anchorage for commercial and pleasure boats.

Materials and Methods

Planning and Environmental Review

A field inspection and feasibility study in 1993 identified the species of rat present and the nontarget species that could be at risk. It confirmed that rat eradication on Kiisgwaii was achievable and recommended the fixed bait-station approach (Taylor & Thomas 1989, 1993) using the anticoagulant poison brodifacoum. The *Canada Environmental Assessment Act* (1995) required the completion of an environmental review and public consul-

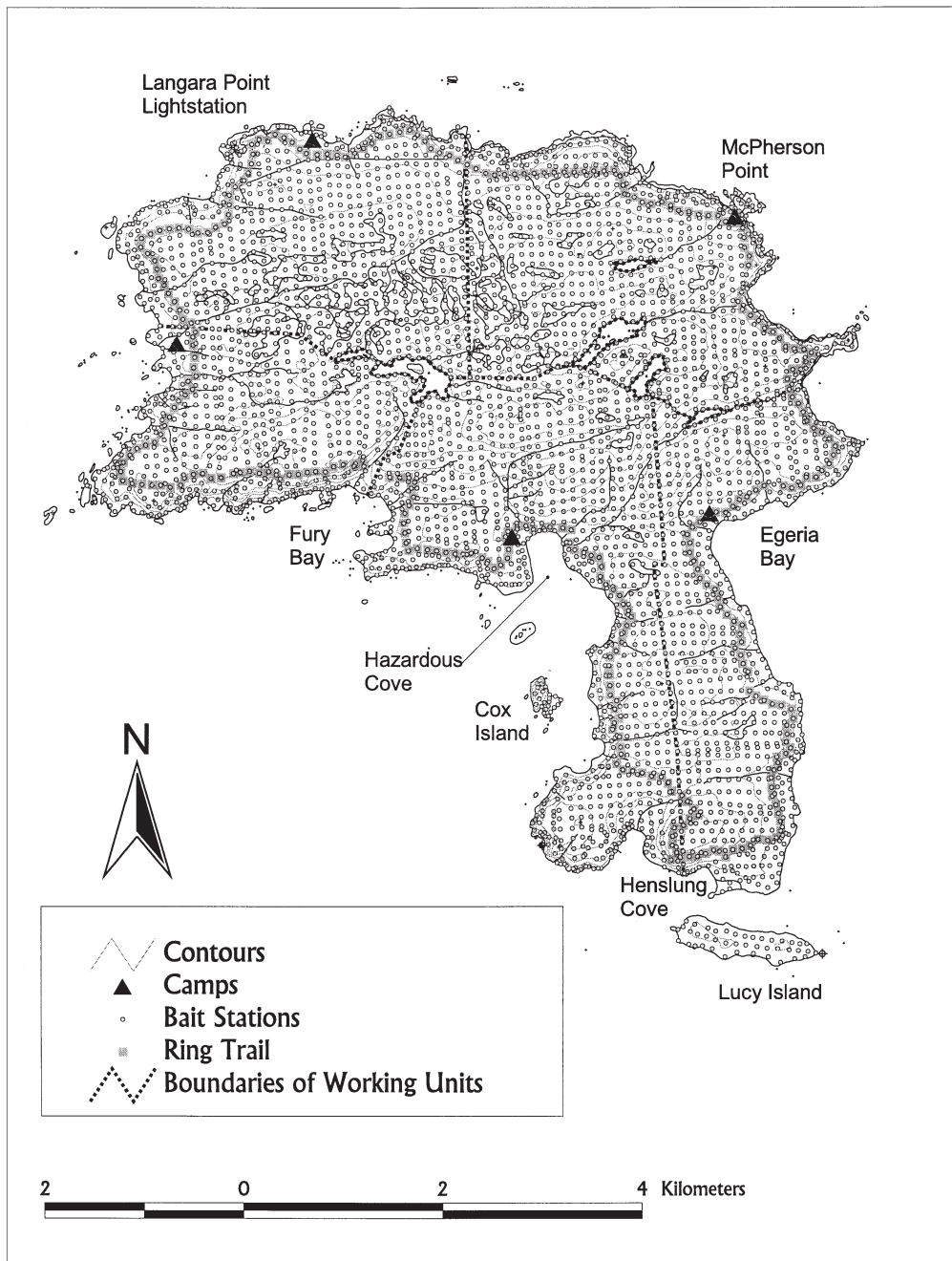


Figure 1. Map of Langara Island with location of bait stations and place names mentioned in the text.

tation. The review addressed secondary and nontarget poisoning, poison persistence, leaching into water systems, preparedness for accidental spillage, and mechanical damage to the environment. The review was attached as an appendix to the feasibility report (Taylor 1993), and copies were distributed to stakeholders and nearby municipalities.

The bait used was a 20-g wax briquette containing brodifacoum at 50 ppm. The formulation is produced and used in New Zealand as TALON 50 WB (Taylor & Thomas 1989, 1993). Environment Canada registered this bait as RATAK+ Weatherblok (ZENECA Agro,

Stoney Creek, Ontario, Canada, L8G 3Z1), specifically for clearing rats from seabird colonies (Kaiser et al. 1997). Brodifacoum acts by blocking the vitamin K₁ oxidation-reduction cycle in the liver, thus preventing the production of blood clotting proteins. Death results from massive internal hemorrhaging (Smith & Greaves 1987). Efficient eradication requires a bait that kills after a single feeding, and only brodifacoum and flocoumafen approach this requirement. Of the two rodenticides, brodifacoum had been much more widely used in successful rodent eradication campaigns in New Zealand. In 1993, little information existed on the toxicity of

brodifacoum to birds and mammals commonly found on Langara Island. However, we assumed it was toxic and relied on bait station design to minimize poisoning of nontarget species.

Bait stations were made from 0.5-m lengths of 100-mm diameter plastic drainage pipe (Fig. 2) secured to the ground with wire hoops. Rats could enter stations through the open sides of the stations and remove baits at will. Baits were checked and replaced via an upper opening (Fig. 2). In addition, we realized that weather could sometimes prevent landings on some offshore islets. We serviced such sites with a simple "hopper" (Fig. 3). Hoppers were made by adding a vertical tube with a cap to the center of the bait station and threading 30 baits onto a rod, which pinned the station to the ground. Rats were not able to carry whole baits from these stations. The unbaited stations were set out 6 weeks before poisoning to circumvent potential problems from the typically neophobic reaction that rats

have with novel structures in their environment (Taylor & Thomas 1989, 1993).

Ecology of Norway Rats

Observations on rats' relative density, distribution, and use of habitat were gathered both before and during the eradication campaign, and included snap-trapping and live-trapping programs across a variety of habitats. Rates of capture are expressed as captures per 100 trap nights (C/100 TN). Rat diet was studied using stomach content analyses (Drever & Harestad 1998).

Lucy Island Pilot Project

We conducted a field trial with toxic baits on Lucy Island in the summer of 1994 to test the method and to ensure that the assumptions in the feasibility study were correct. Bait stations were laid out in May 1994 in



Figure 2. Bait station and poison baits used during eradication campaign. Bait stations were made from 0.5-m lengths of 100-mm diameter plastic drainage pipe, based on the design of Taylor and Thomas (1989, 1993). Rats could enter through either side of the station. Baits were checked and replaced via the upper opening. The opening is circa 120 mm $\times$ 80 mm wide (0.3 the circumference of the pipe). To provide a tight waterproof fit, the cover was two complete rings longer than the opening and more than half the circumference of the pipe.

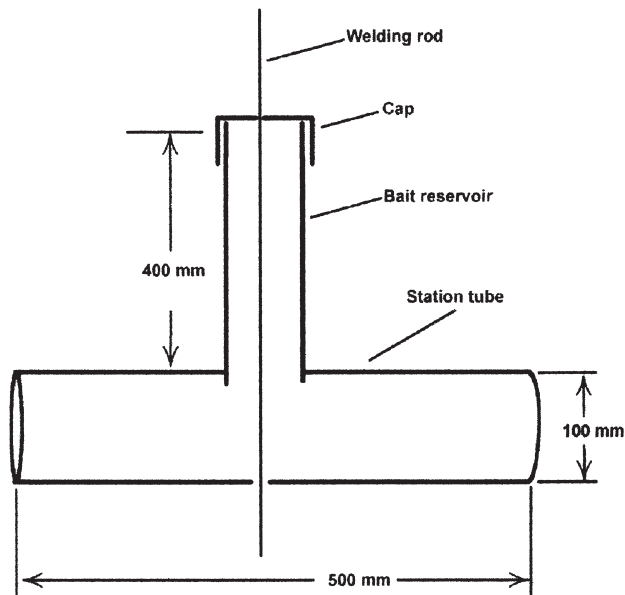


Figure 3. General plan of the hopper used on Langara Island in 1995 where frequent access was not feasible. Rats could enter through either side of the station.

grids that covered the entire island, and baits were placed in stations on 12 July 1994. Stations were checked daily and any missing bait was replaced to maintain a constant bait load of three baits per station. On 21 July 1994, bait loading was increased to four baits per station. On 31 July 1994, baits in each station were placed in protective plastic bags and checked regularly until 17 August 1994, when all baits were removed. Rat and shrew populations were assessed before and after baiting using 49 Longworth live traps on a 1-ha grid and 20 sets of paired snap traps (Howald 1997; Kaiser et al. 1997).

Kiisgwaii Eradication Campaign

The operation required access to each hectare of Langara Island while minimizing environmental impact. When servicing bait stations, operators had to be able to easily find and identify specific locations. We based our initial plans on Forest Cover maps produced by the BC Ministry of Forests. We later had access to digitized orthophoto maps of the Kiisgwaii Archipelago, on which we were able to plot site locations and summarize data.

A walking trail was constructed around Langara Island during August–November 1994. It traversed the western hemlock zone, which offered the easiest route about 200–300 m from the shore and above nearly all seabird-nesting areas. This “ring trail” became the main terrestrial supply and operations route, and was of sufficient standard for easy long-distance travel.

The bait stations were laid out between March–June 1995 on a simple rectilinear grid along flagged routes 100 m apart, with minimum cutting of undergrowth (Fig. 1). We used 100-m spacing between bait stations (1/ha), and set additional stations in coastal areas where rats appeared abundant or the terrain was particularly dissected. Mark and recapture studies in New Zealand had indicated that Norway rats have a home range of approximately 1.2 ha (Moors 1985), and the 100-m spacing assured the presence of at least one station in each home range (Taylor & Thomas, 1989, 1993). Stations were placed on all offshore islets with vegetation even though these were much smaller than 1.0 ha. Each station was marked with an identification number.

For the Kiisgwaii campaign, we divided the archipelago into five convenient working units, each with a camp (Fig. 1), supervisor, and staff. We employed 30 bait station operators who laid baits, recorded the amount taken, and any other activity at the stations. They serviced all stations at 48-hr intervals except for hoppers on offshore islets, which were visited about every 6 days. Management, technical support, and research staff accounted for another 30 persons.

From 10 July to 15 August 1995, we baited the area within 300 m of the shore, where rats were at highest density, with 12 baits/ha every second day. The rest of the island (inside the ring trail) was baited on alternate days with 6 baits/ha. The selected dosage and service frequency was based on New Zealand experience (Taylor & Thomas 1989, 1993; R. H. Taylor, unpublished data) and aimed at 99% control of the rat population over 30 days, and complete eradication in 6 weeks.

We monitored progress of the operation through an electronic database using MAPINFO computer software (Map Info Corporation, Troy, NY). Maps showing active stations and the changing status of the rat population were regularly returned to the crew supervisors. Database files were updated daily from records compiled by the bait station operators.

Assessing Effects on Nontarget Species

Toxic baits may attract species other than rats (primary poisoning of nontarget species) and animals that die after ingesting the bait are likely to be toxic to scavenging species (secondary poisoning). Nontarget poisoning was assessed by (1) collecting and analyzing any bird carcasses (e.g., *Corvus corax* [common ravens]) found during all stages of the eradication (1994–1997) for brodifacoum residues (Howald et al. 2000), (2) analyzing livers of *Melospiza melodia* (song sparrows) and *Corvus caurinus* (northwestern crows) collected between May and August 1995 (Howald 1997), (3) collecting blood samples from adult *Haliaeetus leucocephalus* (bald eagles) between May and August 1995, and analyzing them for

brodifacoum residues and prothrombin times (a screen for deficiencies in clotting factors) (Howald et al. 2000), (4) comparing yearly reproductive success of bald eagles at Langara Island and a reference site on Moresby Island between 1994 and 1996 (Kaiser et al. 1997; Elliott et al. 1998), and (5) comparing trapping rates of shrews in capture-recapture studies and snap-trapping before (May–June 1995) and after (July–August 1995) the initial intensive baiting period (Drever 1997; Howald 1997).

Follow-Up and Monitoring for Rat Activity

We used six snap-trapping lines established in May 1995 to test the efficacy of the eradication campaign (Drever 1997). During August 1995, 144 snap traps baited with a mixture of rolled oats and peanut butter were set for 12 nights (1,728 trap-nights), and 24 apples to detect rat chewing were also set out for 2 nights (48 apple-nights).

For 2 years following the end of the intensive campaign, we monitored for, and safeguarded against, surviving rats by rearming all stations with baits wrapped in plastic bags. This helped prevent deterioration due to water, mold, slugs, snails, or shrews. We checked all stations again in September 1995, the coastal stations in April/May 1996, and all stations in August/September 1996. During these later checks, oiled chew-sticks were placed in groups of two or three at over 100 bait stations and checked for evidence of gnawing by rats. The final check and removal of all flagging, chew-sticks, stations, and bait began in May and ran through August 1997.

Results

Ecology of Norway Rats

Norway rats occurred throughout much of Langara Island. Capture rates differed significantly between the coast (mean $\pm$ SE = 16.9 ± 3.8 C/100 TN) and the interior (4.8 ± 0.8 C/100 TN [Student-Newman-Keuls test, $p < 0.05$; Drever 1997]). None was caught in the upland bogs. Overall trapping results indicated that rats were three times as dense in coastal habitat as in the interior of the island (Drever 1997; Kaiser et al. 1997).

Norway rats on Langara Island fed omnivorously on a wide assortment of food types (Fig. 4). Foods occurring with high frequency and high volumes in rat stomachs ($>50\%$) were plant shoots, salal berries, tissues of ancient murrelets, amphipods, Sitka spruce seeds, and fungi (Drever & Harestad 1998). Tissues from ancient murrelets occurred with the highest frequency (53% of stomachs examined, $n = 19$) and volume (mean percent volume of stomach: mean = $41 \pm 11\%$ [SE]) in diets of rats trapped in the murrelet colony. We found caches of ancient murrelet chicks and eggs with signs of rat predation, as well

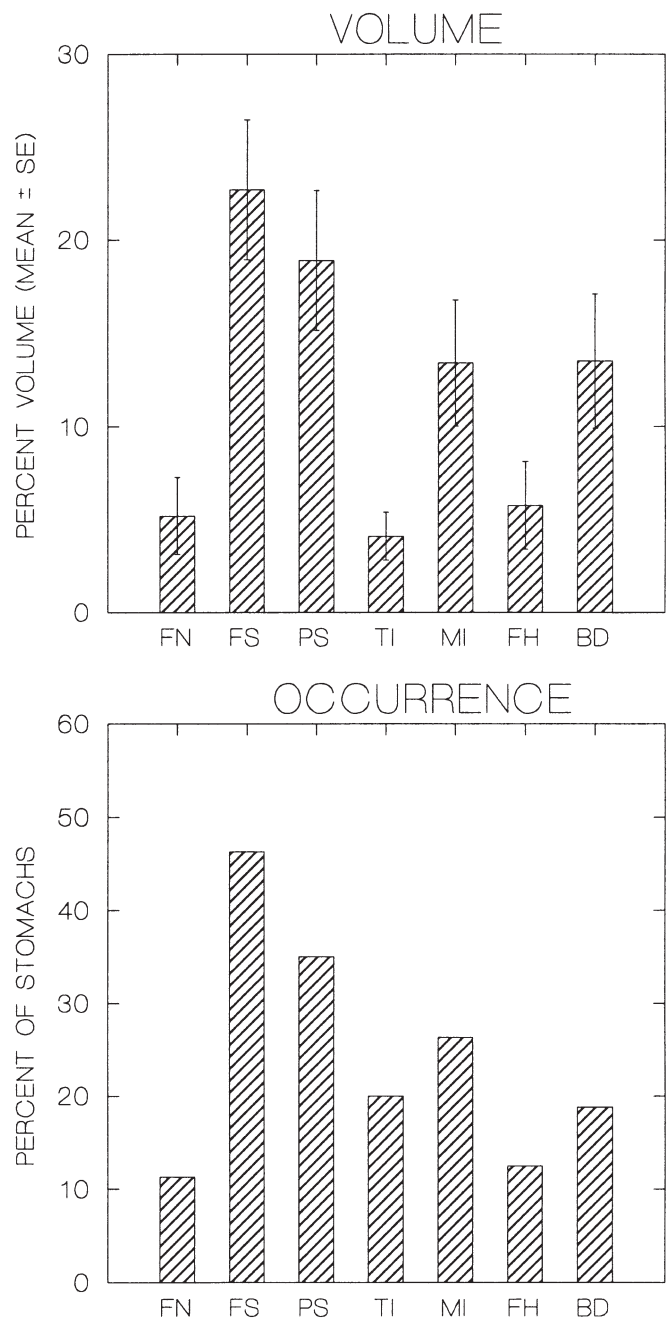


Figure 4. Percent occurrence and percent volume of food groups in stomachs of 80 Norway rats trapped between April and July 1995 on Langara Island, British Columbia. Food groups are: FN, fungi; FS, fruits and seeds; PS, plant shoots; TI, terrestrial invertebrates; MI, marine invertebrates; FH, fish; BD, birds (from Drever 1997).

as dead adult ancient murrelets, indicating that rats preyed upon all life stages of ancient murrelets. Plant shoots and terrestrial invertebrates were most frequent in rats from the interior, whereas marine invertebrates and fruits and seeds were most common in stomach contents from rats from coastal areas (Drever & Harestad 1998).

Lucy Island Pilot Project

Stations became active, that is, rats began taking baits, on 13 July 1994 (day 2 of the trial on Lucy Island). Activity built to a peak on 29 July 1994 and had ceased by 5 August 1994. Rats took bait from 74% of the stations and shrews were active at 28% of the stations. Capture rate of shrews in the 1-ha grid dropped from 19.9 C/100 TN in July 1994 to 2.8 C/100 TN in August 1994. One juvenile Norway rat was trapped after the baiting (18 August 1994), and a second carcass was found in a lost trap in May 1995. An additional rat was trapped in April 1995, but it was not clear whether this was a survivor or immigrant that had swum the narrow channel from Langara Island (Drever 1997). The bait stations were rebaited in July 1995 and treated as part of the main eradication campaign during which several stations showed signs of rat activity.

Kiisgwaii Eradication Campaign

We began placing bait in the first of the 3,905 bait stations on 10 July 1995. A total of 30,577 baits were placed in stations between July 10 and 12, 1995. Rats began taking baits within a few hours of placement, and 361 stations were active by 14 July 1995. By 27 July 1995, rats had taken 95% of the total number of baits they would take during the entire campaign. By July 31 1995, only 14 stations were active. Small pockets of activity continued until 9 August 1995 (31 days after baits were first placed into stations). There was no rat activity at bait stations on Kiisgwaii between 9 August and 15 August 1995. Rats took a total of 14,500 baits. Many more baits were replaced due to the effects of shrews, slugs, and mold.

Stations around the coast were more likely to become active and remain active for longer than interior stations (Table 1). On average, stations experienced their first rat activity after a lag time of 6.5 days. Lag time between initial bait placement and start of activity did not differ significantly between coastal and interior stations (Mann-Whitney $U = 134431.0$, $p = 0.693$), although it had a significant negative relationship with duration of activity (Spearman $R = -0.21$, $n = 1080$, $p < 0.0001$).

Table 1. Activity at coastal and interior bait stations in Kiisgwaii during the 1995 eradication campaign.

	Coast	Interior
Total number of bait stations	1293	2514
Number of active bait stations	661	410
Percent of active bait stations	51	16
Mean lag time from arming to activity (days)	6.7	6.5
Mean duration of activity (days)	3.9 ^a	2.1 ^b

^{a,b} Significantly different: Mann-Whitney $U = 84826.5$, $p < 0.0001$. Activity is defined as removal of ≥ 1 bait block by rats from bait station (from Kaiser et al. 1997).

Assessing Effects on Nontarget Species

The loss of ravens through secondary poisoning appeared small during the eradication operation of July and August 1995, but primary poisoning became a more obvious problem between September 1995 and May 1996 during the follow-up and monitoring phase (Kaiser et al. 1997). Circumstantial evidence exists that ravens tampered with bait stations from the start of the eradication campaign—lids were occasionally found to be missing and puzzling patterns appeared in bait station activity in areas where there were known to be few rats. This disturbance increased following the main poisoning campaign. We found other evidence that ravens were raiding bait stations. Twelve pellets regurgitated by ravens were found which consisted largely of blue wax from bait blocks and fragments of the plastic bags that wrapped baits. The remains of thirteen ravens were collected from various parts of the island, including eight that had died of pulmonary hemorrhaging (Howald et al. 2000). There were still ravens present throughout the summers of 1996 and 1997, but numbers seemed low in comparison with previous years and to their abundance on nearby Graham Island. None of the five known nest sites was active.

Brodifacoum residues were detected in northwestern crows, song sparrows, and blood samples of bald eagles (Howald 1997), but these species suffered no detectable population decline. A few bald eagles likely gained access to the poison by scavenging raven carcasses. Bald eagle nest occupancy was similar at Langara Island in 1995 and 1996 (Table 2), indicating no immediate impact on the local breeding population. Breeding success of eagles on Langara Island in 1996 was significantly lower than in previous years, as it was at the reference site, indicating some phenomenon that affected both sites equally (Table 2). Elliott et al. (1998) speculate that reduced food supply in the marine environment resulted in lowered breeding success at both sites. Snap-trapping rates of shrews decreased significantly from 5.8 ± 1.1 (SE) C/100 TN to 2.0 ± 0.9 (SE) after the eradication campaign (Student-Newman-Keuls test, $P < 0.05$; Drever 1997). However, this provides equivocal proof of a poisoning effect since the trapping itself removes animals from the population. No significant decrease in trapping rate occurred at the live traps (Howald 1997).

Follow-Up and Monitoring for Rat Activity

No rats were caught during the two snap trapping sessions in August 1995, and there were no incisor marks of rats on the apple indicator baits. When all the stations were checked in September 1995, seven stations near the fishing lodges showed activity by rats. These were rearmed and a further 17 baits taken. We found no

Table 2. Nest success and production of young for bald eagles at Langara Island and at a reference site in the South Moresby area (from Kaiser et al. 1997).

Study Area	Year	No. of Occupied Territories	No. of Successful Nests	% Nest Success	No. of Young Produced	Young/Occupied Nest
South Moresby	1994	19	5	26	6	0.32
	1995	17	6	39	6	0.35
	1996	20	2	10	2	0.10
	Mean	19 ± 2	4 ± 2	25 ± 15	5 ± 2	0.26 ± 0.17 ^a
Langara Island	1994	22	13	59	16	0.73
	1995	21	12	57	19	0.83
	1996	20	7	35	10	0.50
	Mean	21 ± 1	11 ± 3	50 ± 13	15 ± 5	0.69 ± 0.17 ^b

^{a,b} Overall by location, means that do not share the same letter are significantly different ($p < 0.05$).

sign of rat activity there in October 1995, or in May 1996 or 1997. Five months after the extermination campaign, in January 1996, the light keeper at Langara Point trapped an adult male Norway rat in a grain store. Eight additional stations were placed around the lighthouse. No rat activity was found in subsequent checks at the lighthouse.

During the April–May check in 1996, we found 52 stations disturbed by ravens. No firm evidence of the continued presence of rats was found. The summer 1996 check of all stations showed continuing disturbance by ravens, but there was no sign of rats having disturbed the bait or chew-sticks or of their feeding sign, tracks and runways in bird colonies, on sandy beaches, or through ground vegetation.

There was no evidence of surviving rats from the final bait station and chew-stick checks in spring and summer 1997. Inspections in May 1997 of the Egeria Bay shoreline, the ancient murrelet colony at McPherson Point, the lighthouse, Fury Bay, Hazardous and Henslung coves, and Cox and Lucy islands failed to find rat feces, foot prints, or signs of predation by rats.

Discussion

The extermination proceeded according to plan and there was no rat activity at the bait stations at the end of the main campaign in August 1995. Two small pockets of rats survived in areas of human habitation, but these were quickly identified during the programmed station checks and eliminated. Ancient murrelets bred in the colony in May 1996 and 1997, with no signs of disturbance by rats.

Rats took bait from about 25% of the stations and all other evidence suggests that only about one quarter of Langara Island, mostly coastal areas, supported rats. In total rats took approximately 14,500 baits or about 90 g/ha. A comparison with the consumption of 1,185 g/ha on Breaksea Island, New Zealand (Taylor & Thomas 1993), where the level of baiting was similar to that

employed on Langara Island, suggests that the average density of rats for the whole of Langara was only 7.6% of that on Breaksea Island. From snap-trapping, Taylor and Thomas (1993) calculated the density on Breaksea Island to be 13 rats/ha before baiting. By interpolation, the average density of rats on Langara Island would have been about 1.0 rats/ha. This estimate of density is supported by the results of trapping on Langara Island in May–June 1995. A mark-recapture study in the coastal zone near McPherson Point indicated that about 45 rats occupied 15 ha (Drever 1997). Snap trapping also indicated about 3 rats/ha in the coastal zone and about 1 rat/ha overall (16.9 C/100 TN along the coast and 4.8 C/100 TN in the interior) (Drever 1997). Thus, a coarse estimate of the total rat population of Langara Islands would be about 3,000 before poisoning.

Analysis of rat stomach contents showed that the diet of rats varied around the island, and that they were exploiting the highest quality and most readily available food items in each habitat (Drever & Harestad 1998). Rats in the ancient murrelet colony consumed primarily ancient murrelets. Subsequent analyses of stable isotope concentrations in liver and muscle tissues of rats later demonstrated that stomach contents underestimated the true importance of murrelets in the diets of rats (Hobson et al. 1999). The incidence of plant shoots in the diet increased with distance from the sea. Terrestrial invertebrates, which are higher in protein and have a greater variety of essential amino acids than most plants (Needham 1964; Clark 1981; Moors 1985), were consumed in greater frequencies and volumes in the interior habitats. Near the coast, amphipods and other marine invertebrates provide much of this protein. Several species of amphipods are nocturnal, feed in large masses on vegetation at or near the high-tide line on exposed beaches (Kozloff 1993), and thus are available when rats are most active.

The impact of the campaign on common ravens was greater than expected, but not catastrophic. Ravens were secondarily poisoned from scavenging rat car-

casses and primarily from raiding bait stations. Probably 50% or more of the local raven population was killed. Immigrants from the much larger population on Graham Island appear to be already replacing those poisoned on Langara Island.

The campaign undoubtedly led to the death of many shrews but the population effects were apparently short lived. Shrews have likely benefited from the absence of rats through reduced competition for insect prey and possibly decreased predation.

Management Implications

Methodology

The technique proven on Breaksea Island (Taylor & Thomas 1993) was effectively extrapolated to a much larger island. The duration of eradication campaigns is related to bait availability in an exponential decline, such that higher bait availability results in faster rat mortality (Fig. 5). The Langara Island campaign followed this predicted curve. The extrapolation was straightforward and did not require any significant changes to the protocol or to the methodology. The rat eradication on Langara Island demonstrated the inherent robustness of the method in that it was carried out by seasonal laborers instead of wildlife technicians supported by dedicated volunteers, and that it succeeded in spite of an inability to carry out monitoring checks with the same frequency as the New Zealand campaigns.

The dietary plasticity of the Norway rat underscores the importance of using very attractive and proven bait over an entire island when eradication is being at-

tempted. Long-term coastal baiting alone will not eradicate rats because the inland habitat will sustain rat populations at low densities. The importance of seeds and shoots in the diet of Norway rats implies that a poison campaign during winter, when these foods are least available, would maximize bait uptake (Taylor & Thomas 1993). However, stormy weather in the northeast Pacific during winter creates a very high hazard to humans, and the large manpower needs of this project precluded a winter campaign.

Nontarget Species

Our inability to reduce the threat to ravens stems from the large area involved in this project and our inability to respond effectively in winter. The method can easily be improved to reduce that risk in future efforts by changing the timing of the operation and modifying the bait stations. For instance, ravens would be unable to take baits from longer stations or s-shaped stations securely anchored to the ground and with secure lids.

Possible Arrival of New Rats

Boat traffic and barges servicing the fishing lodges on Langara Island present a risk of reintroducing rats onto the island, and such traffic will undoubtedly increase as the area becomes more developed. Fishing lodges have voluntarily treated their barges by professional pest management companies before being brought to Langara Island, and lodge staff are sensitized to the need to keep the island rat free. Funds are currently sought to establish a monitoring and response system using lodge personnel. The status of the ancient murrelet colony and the possible recolonization by extirpated species such as storm petrels will be evaluated in 1999. This will be the definitive measure of the success of the eradication campaign.

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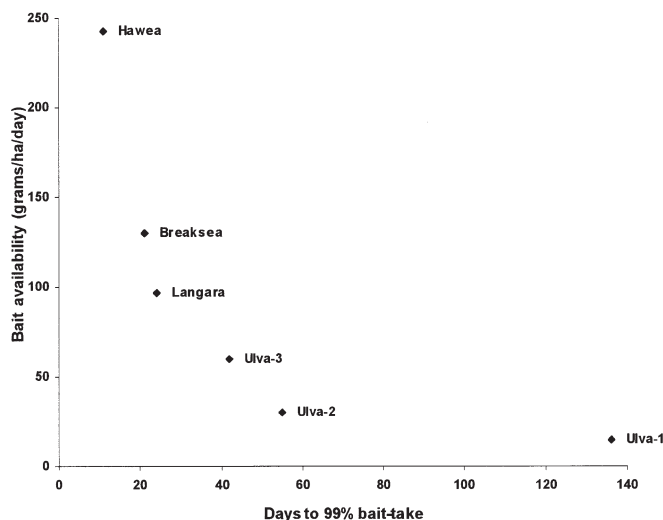


Figure 5. Relationship between bait availability and duration of rat-eradication campaigns at Langara Island and five islands in New Zealand (after Taylor and Thomas 1989, 1993; R. H. Taylor, unpublished data).

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